Learned models of biology: atlases, foundation models, and the honest scorecard

- Biyolojinin öğrenilmiş modelleri: atlaslar, temel modeller ve dürüst karne
- BIB Week 11

Ad of the Week #11

- Skills Ledger: tick public reference atlases and benchmarking against a baseline

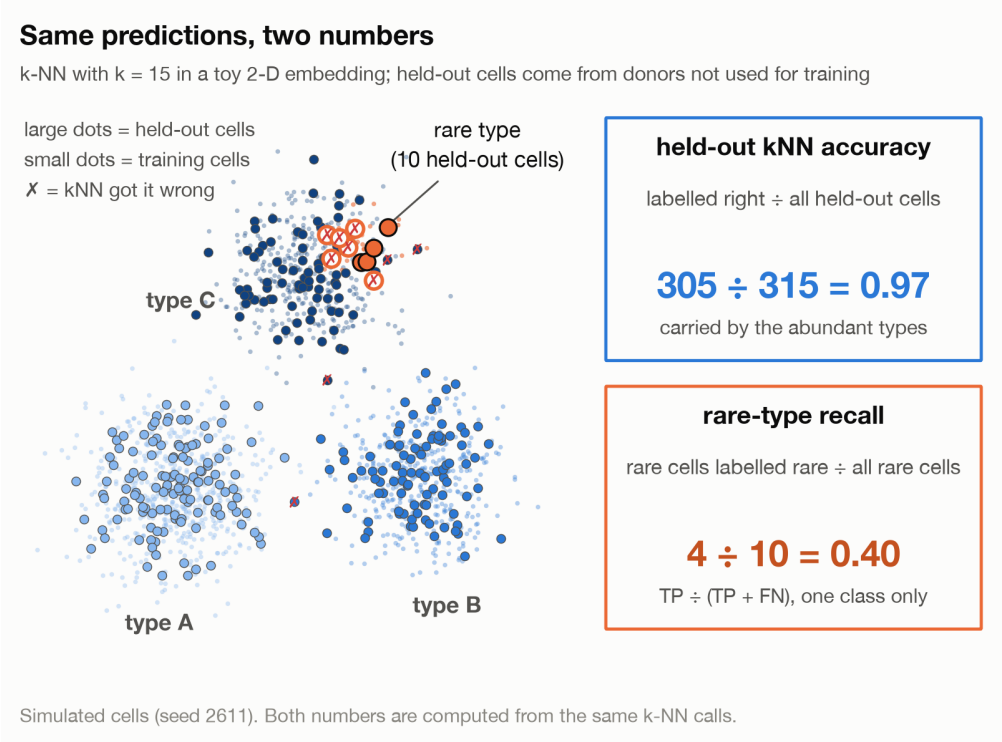
The rule, back from Week 9

- Week 9 put one sentence on its own slide and promised it would come back. It is today's whole argument.
- A benchmark (kıyaslama) score means nothing until you know what the baseline (temel çizgi) scores
- Strong baseline: a simple method done properly, on the same data — not a straw man chosen to lose
- Simple: the majority class, the mean, plain PCA, adding two effects together
- Today: atlases → foundation models (temel modeller) → the scorecard → laundering → where they win

A benchmark without a strong simple baseline is not a benchmark.

Two numbers you already own from Week 9

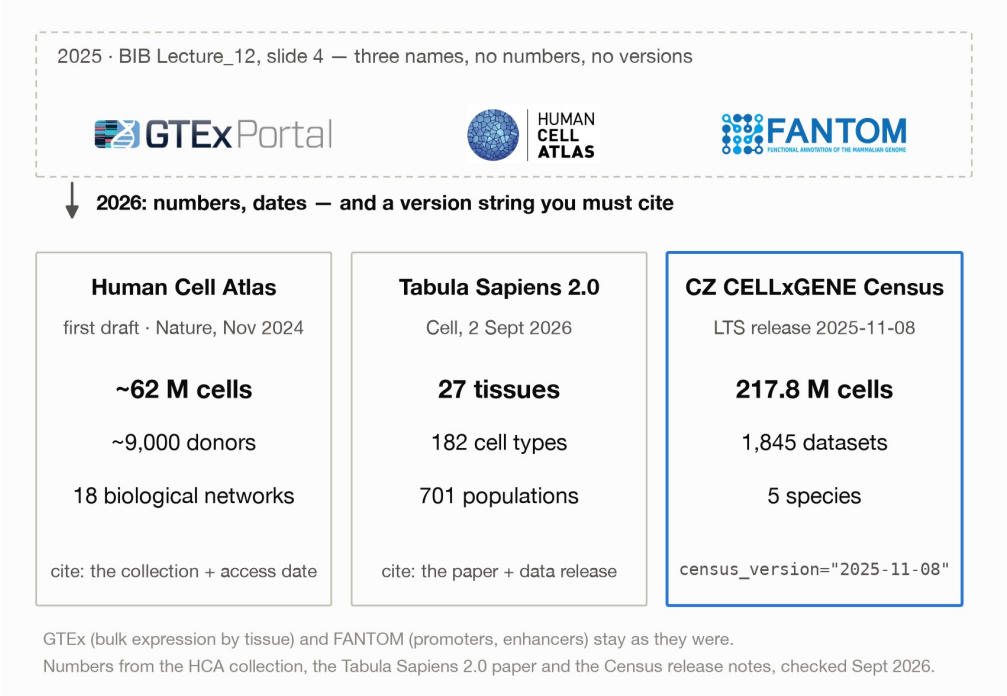
- Held-out kNN accuracy (k-NN doğruluğu): label each held-out cell by vote of its k nearest training cells
  - accuracy = labelled right ÷ all held-out cells
- Rare-type retrieval recall (nadir tip duyarlılığı): TP ÷ (TP + FN), counted for the rare type only
- Both are computed inside an embedding (gömme): change the coordinate system and the neighbours change
- Held out by donor, not by cell — the unit of independence from Weeks 6, 9 and 10
- Accuracy is carried by the abundant types (the base rate, Week 9). Rare-type recall is not.



Simulated: the same k-NN calls give accuracy 0.97 and rare-type recall 0.40

Atlases: from three logos to an access layer

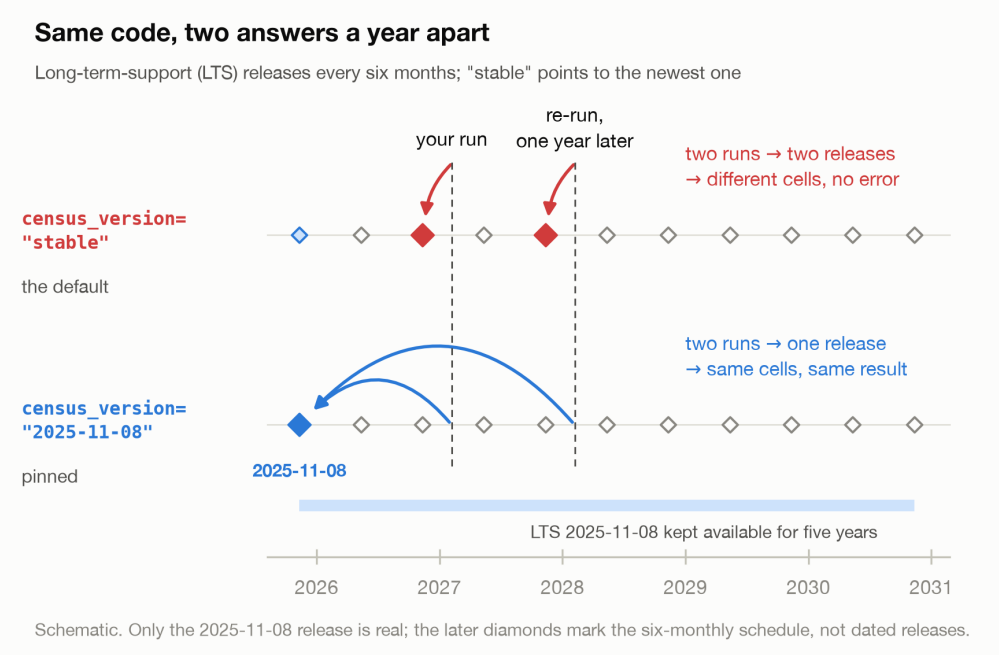
- 2025's slide named GTEx (bulk expression by tissue), FANTOM (promoters, enhancers) and HCA
- Human Cell Atlas first draft (Nature, Nov 2024): ~62 M cells, ~9,000 donors, 18 biological networks
- Tabula Sapiens 2.0 (Cell, 2 Sept 2026): 27 tissues, 182 ontology-controlled cell types, 701 populations
- CZ CELLxGENE: browse it at cellxgene.cziscience.com; query it from code through the Census
- A reference atlas (atlas) is something you query and cite, not a picture you admire



BIB Lecture_12 (2025) slide 4, updated: the same three consortia, then the 2026 numbers and what to cite

The release string is part of your result

- CZ CELLxGENE Census LTS release 2025-11-08: 1,845 datasets, 217.8 M cells, five species
- A new long-term-support (LTS) release every six months; each one kept available for five years
- Reference data is now a versioned public API (sürümlü açık API): your code asks, a release answers
- Leave the version out and you get "stable": whichever release is newest on the day you run
- Week 1's rule: resource, release, access date. The release string goes in GENOMES G and S.

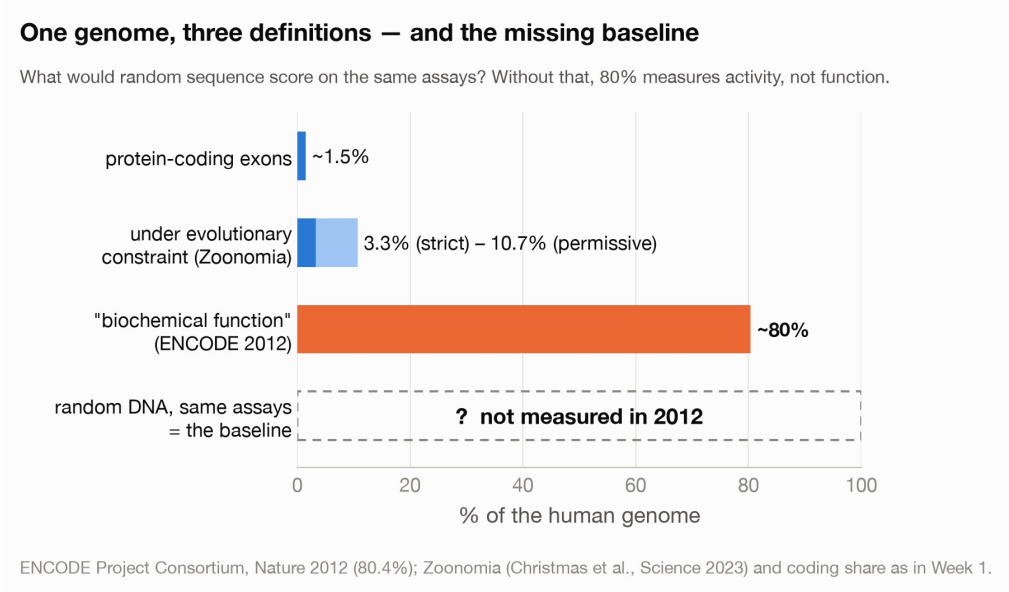


Schematic: unpinned code reads a different release each time it runs; pinned code reads the same one

```
census = cellxgene_census.open_soma(
    census_version="2025-11-08")
```

The ENCODE debate, as promised in Week 1

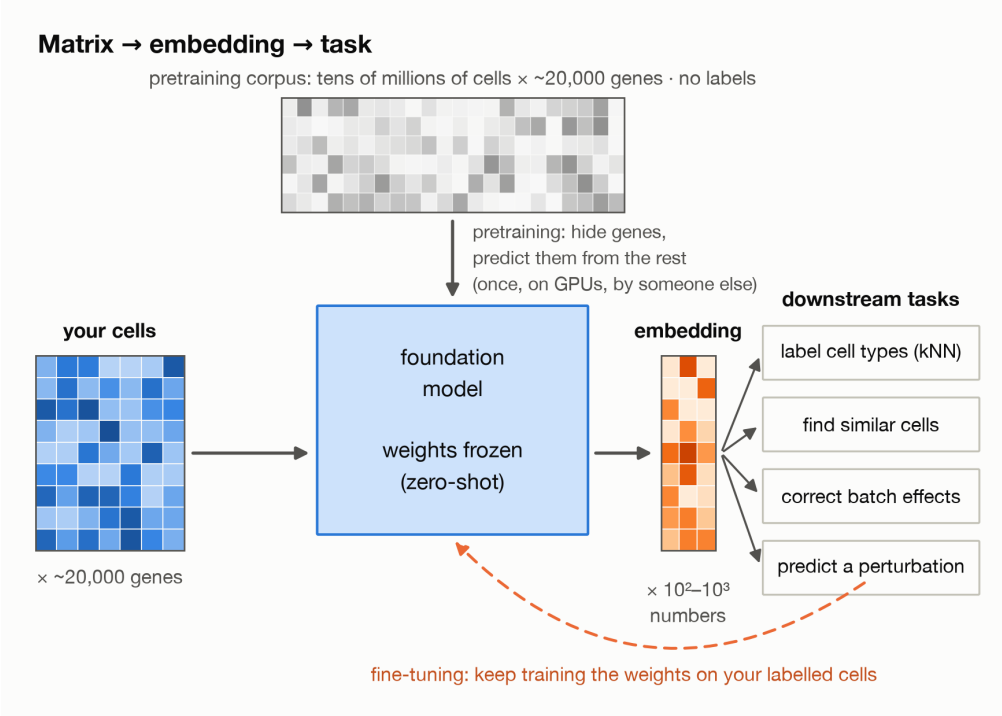
- 2012: ENCODE reported a "biochemical function" for ~80% of the human genome
- "Function" meant a reproducible biochemical signature: transcribed, bound or marked somewhere
- Evolutionary constraint (evrimsel kısıt) says ~3–11% (Week 1). Same genome, an order of magnitude apart.
- The critics' question: how active is random DNA in the same assays? The 2012 headline had no answer.
- Without that baseline, 80% measures activity, not function. The claim was walked back.



One genome, three definitions — and the baseline nobody measured

What a foundation model is, in words you own

- Foundation model (temel model): pretrained (ön eğitim) on tens of millions of unlabelled cells
- Pretraining: hide genes in each cell, predict them from the rest. The hidden genes are the answer key.
- What you take home is an embedding: each cell becomes 10^2–10^3 numbers instead of ~20,000 genes
- Downstream task (alt görev): label cells, find similar cells, correct batches, predict perturbations
- Zero-shot (sıfır atış): use it as it comes. Fine-tuning (ince ayar): train it further on your labels
- Named once: Geneformer, scGPT, scFoundation, UCE, SCimilarity, TranscriptFormer



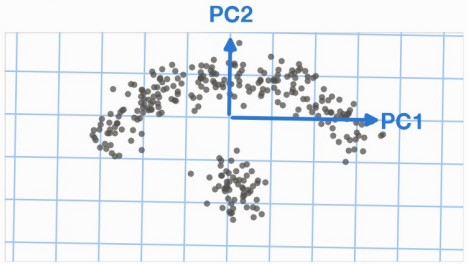
Matrix → embedding → downstream task; pretraining happens once, somewhere else

An embedding is a learned coordinate system

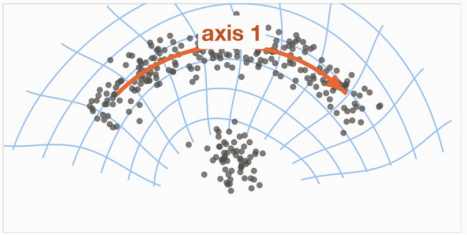
- PCA (Week 9) is a hand-made coordinate system (koordinat sistemi): a fixed rule, applied to your cells
- An embedding is a learned one: its axes were shaped by tens of millions of other cells
- A PCA axis is readable — a weighted sum of genes. An embedding axis is not, one by one.
- PCA finds the biggest variance in your data, which may be batch (Week 9) rather than biology
- An embedding brings outside knowledge, and the biases of whatever it was trained on
- Week 5: an embedding of genomic data is genomic data. It is not de-identified.

Two coordinate systems over the same cells

PCA — hand-made
straight axes from a fixed rule



Embedding — learned
axes bent by what the network learned

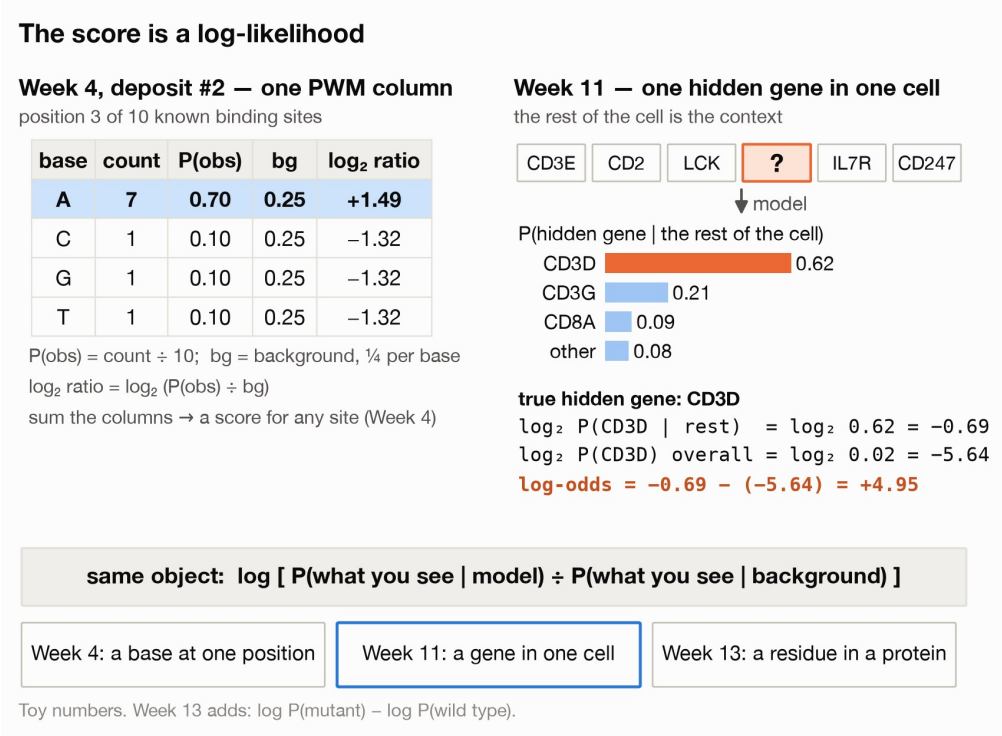


	PCA (Week 9)	learned embedding
built from	your cells only	tens of millions of other cells
the rule	maximum variance, at right angles	whatever made hidden genes predictable
read one axis?	yes — a weighted sum of genes	no, not one by one
cost	seconds on a laptop	GPU-weeks or more, paid once by others

Simulated cells: straight axes from a rule, curved axes learned from other data

The score is a log-likelihood: Week 4, withdrawn

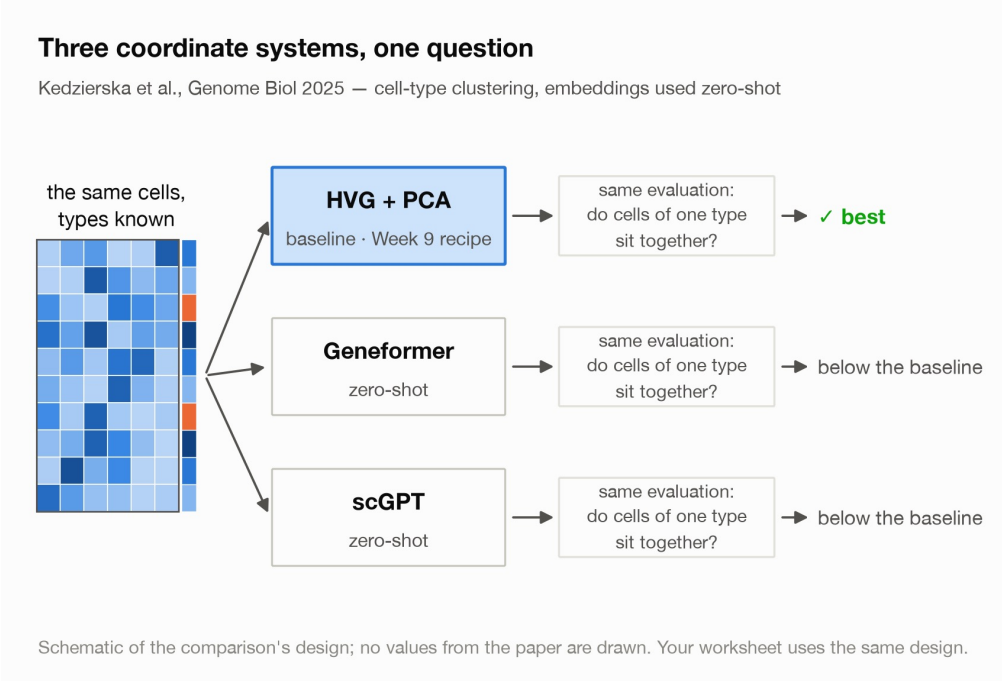
- Week 4 deposit #2: the PWM (konum ağırlık matrisi) — per position, log(observed ÷ background)
- A masked-gene model does the same with one change: the column depends on the rest of the cell
- Score of a hidden gene = log P(gene | rest of the cell), a log-likelihood (log-olabilirlik)
- Subtract the background, log P(gene) over all cells, and you have Week 4's log-odds exactly
- Pretraining pushes that number up, averaged over every hidden gene
- Third meeting, Week 13: log P(mutant) − log P(wild type), from a protein model



Toy numbers: one PWM column beside one hidden gene — the same log-odds object

Scorecard 1: zero-shot embeddings versus plain PCA

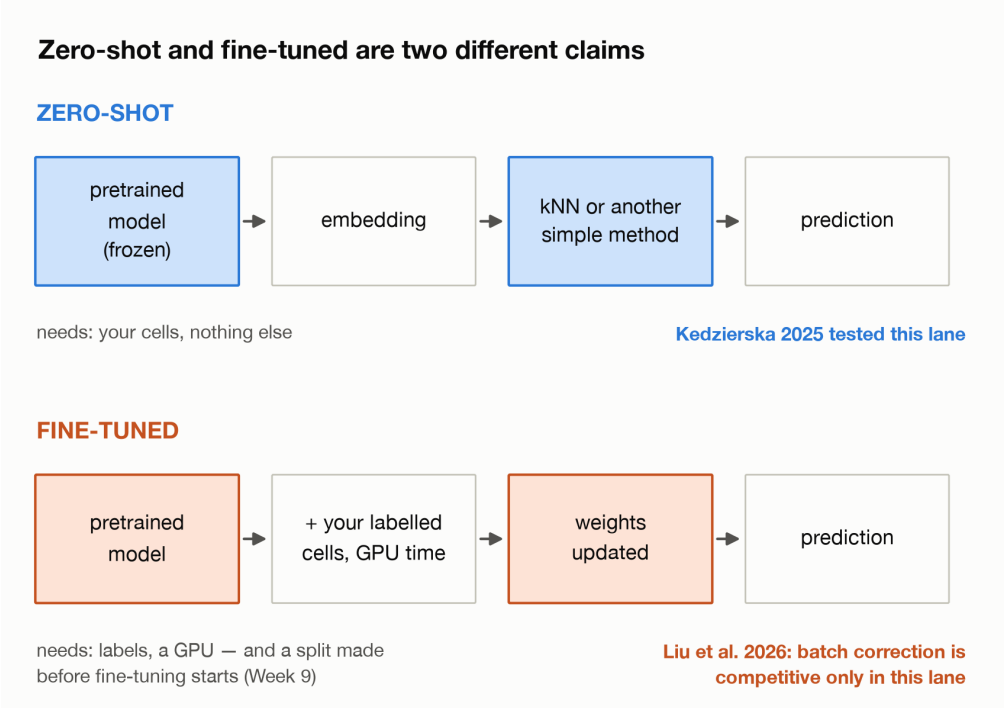
- Kedzierska et al., Genome Biol 2025: Geneformer and scGPT embeddings used zero-shot
- Task: cell-type clustering — do cells of one type end up together?
- Baseline: highly variable genes (yüksek değişkenlikli genler, HVG) + PCA, seconds on a laptop
- Result: the plain baseline separated cell types better than both foundation models
- Not "foundation models fail" — "these zero-shot embeddings did not beat PCA at this task"
- Your worksheet repeats the design: HVG + PCA-50 versus scGPT versus Geneformer, on the same cells



Schematic of the comparison: the same cells, three coordinate systems, one evaluation

Scorecard 2: more tasks, and what fine-tuning costs

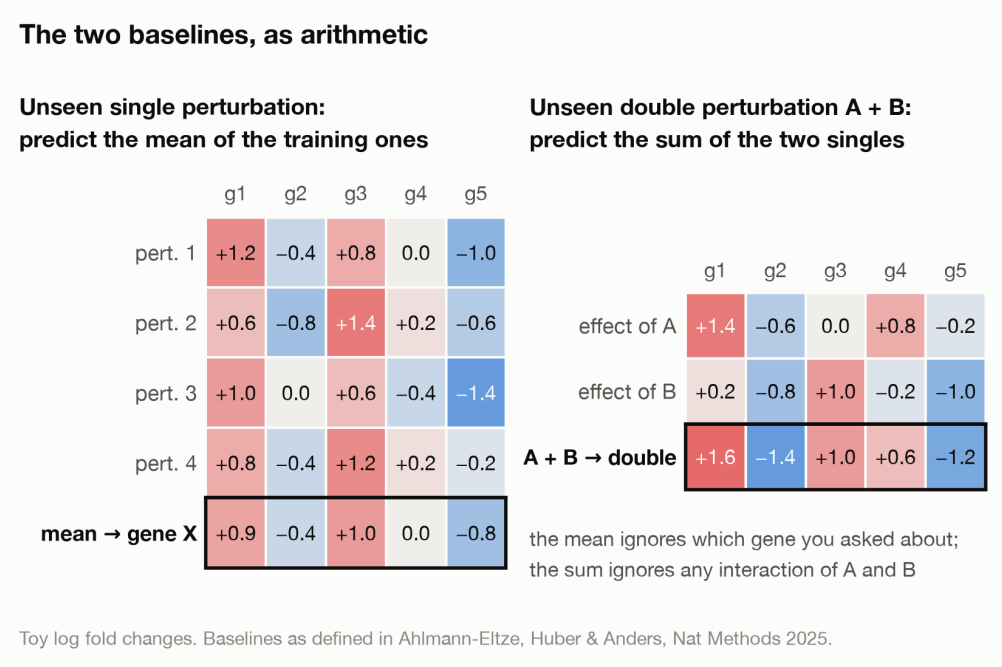
- scEval: ten foundation models, eight tasks — no consistent win over task-specific methods
- Task-specific method: a tool built for one job, such as a dedicated batch-correction method
- Liu et al., Adv Sci 2026: batch correction (parti etkisi düzeltme) is competitive only after fine-tuning
- Fine-tuning costs labelled cells, GPU time, and a fresh chance to leak (Week 9)
- "Zero-shot" and "fine-tuned" are different claims. A benchmark must say which one it tested.



Zero-shot and fine-tuned use of the same pretrained model, and what each one needs

The flagship: predicting a perturbation nobody ran

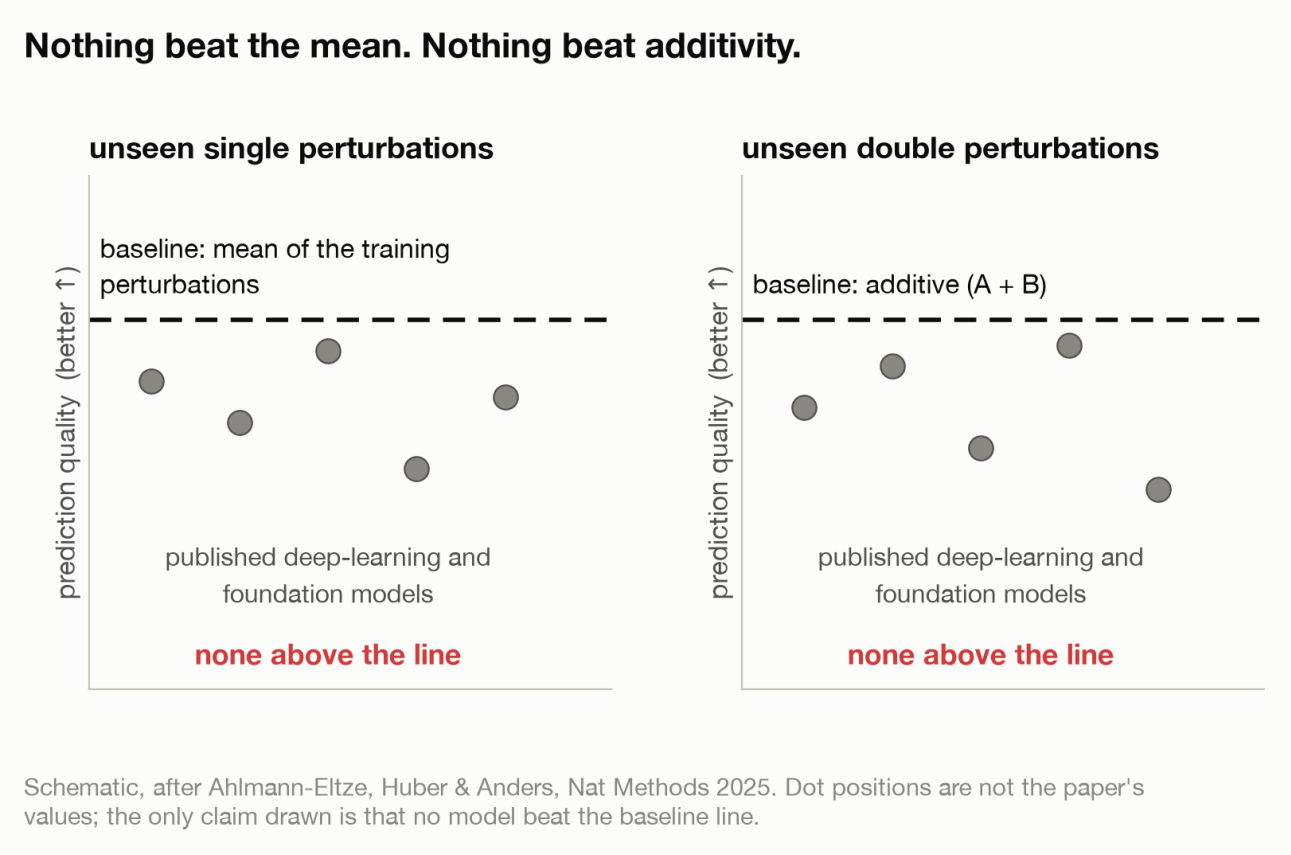
- Task: switch one gene up or down with CRISPR; predict how every other gene's expression changes
- The hard version: the gene was never perturbed in training, or the pair never seen together
- Baseline 1, unseen single: predict the mean of all the training perturbations
- Baseline 2, unseen double A + B: predict effect(A) + effect(B) — additivity (toplamsallık)
- Neither knows any biology: one ignores which gene you asked about, the other ignores interaction
- Ahlmann-Eltze, Huber & Anders, Nature Methods 2025



Toy numbers: the two baselines are a mean and a sum

Nothing beat the mean. Nothing beat additivity.

- Unseen single perturbations: no model beat predicting the mean of the training perturbations
- Unseen double perturbations: no model beat additivity
- Not "the models are useless": nobody can tell until the baseline is on the same axis
- Before you trust a "virtual cell": what does the mean score on the same split?



Schematic, after Ahlmann-Eltze, Huber & Anders, Nat Methods 2025: no model above the baseline line

Virtual Cell Challenge: the organisers' verdict

- Virtual Cell Challenge 2025 (Arc Institute): predict held-out perturbations
- Organisers' wrap-up: models not yet consistently beating naive baselines
- Most entrants did worse than the baseline on mean absolute error (ortalama mutlak hata)
- 2026 zero-shot edition closes 5 November 2026, inside this semester

The honest scorecard: five results, one direction

study	task	compared against	verdict
Kedzierska et al. Genome Biol 2025	cell-type clustering, zero-shot	HVG + PCA	**baseline won**
scEval	8 tasks × 10 models	task-specific methods	**no consistent win**
Liu et al. Adv Sci 2026	batch correction	established methods	**competitive only after fine-tuning**
Ahlmann-Eltze, Huber & Anders, Nat Methods 2025	unseen single and double perturbations	mean; additivity	**baseline not beaten**
Virtual Cell Challenge 2025 (Arc Institute)	perturbation response	naive baseline	**most entrants worse on MAE**

Each verdict as its source states it; no effect sizes are drawn.

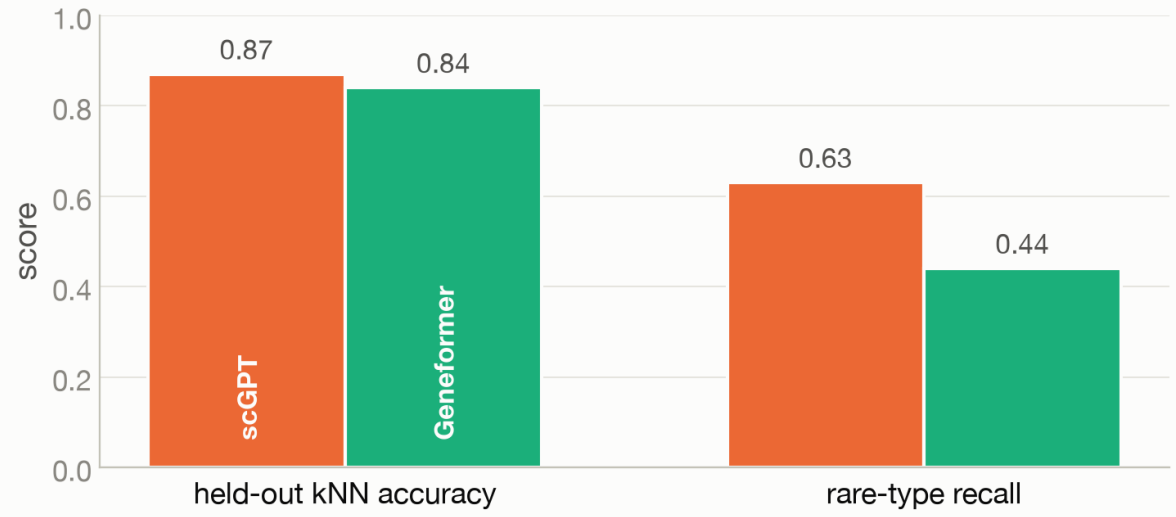
The honest scorecard: each source's own verdict, no effect sizes

Read this result the way a reader would

- Two foundation models, two metrics, one bar chart — as it might appear in a paper or a press release
- Write down the one sentence you would conclude from it
- Nothing on this slide is false: every number matches the full table on the next slide

Zero-shot foundation-model embeddings, held-out cells

embedding	held-out kNN accuracy	rare-type recall
scGPT, zero-shot	0.87	0.63
Geneformer, zero-shot	0.84	0.44



Illustrative numbers — the worksheet carries the real run.

Illustrative numbers — the worksheet carries the real run

Benchmark laundering: the row someone deleted

- Benchmark laundering (kıyaslama aklama): every number true, the conclusion false — by omission
- 1. Baseline: is the strongest simple method in the table, run on the same split?
- 2. Split: what is the independent unit, and had pretraining already seen the test cells?
- 3. Metric: who chose it, and does the winner change under another one?
- 4. Budget: zero-shot or fine-tuned, and did the baseline get the same tuning effort?
- 5. Stake: who ran it, and what do they gain if the model wins? (Week 13)

The same table, with its baseline row

embedding	held-out kNN accuracy	rare-type recall
HVG + PCA-50 (baseline)	**0.91 ✓ best**	**0.48**
scGPT, zero-shot	0.87	**0.63 ✓ best**
Geneformer, zero-shot	0.84	0.44

↑ the blue row is the one deleted on the previous slide

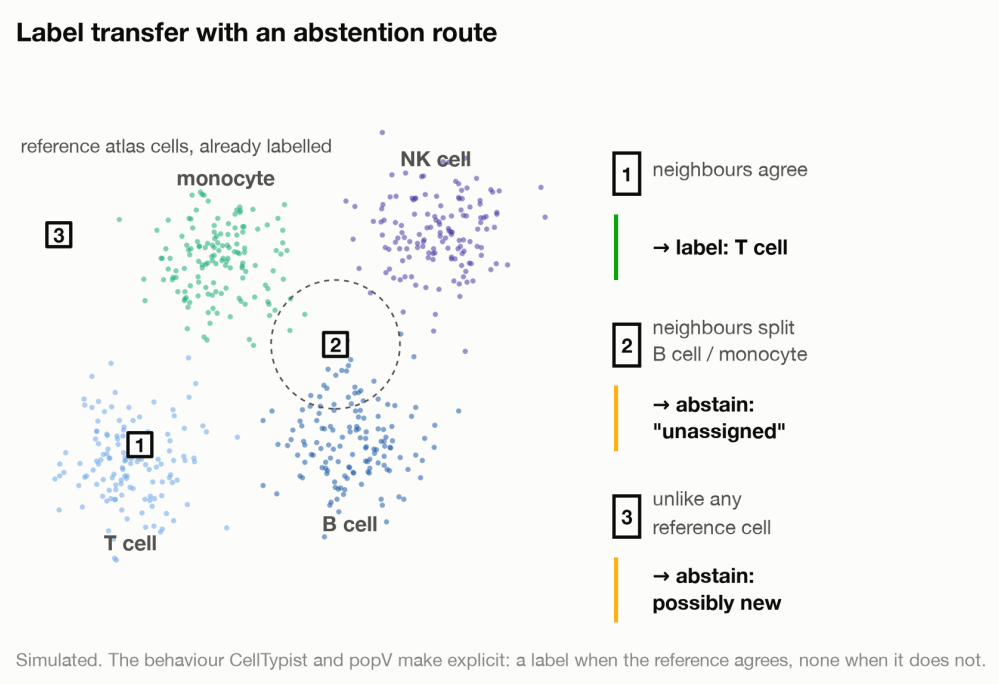
The winner changes with the metric. Both facts go in the report.

Illustrative numbers — the worksheet carries the real run.

Illustrative numbers — the worksheet carries the real run

Where the model genuinely wins

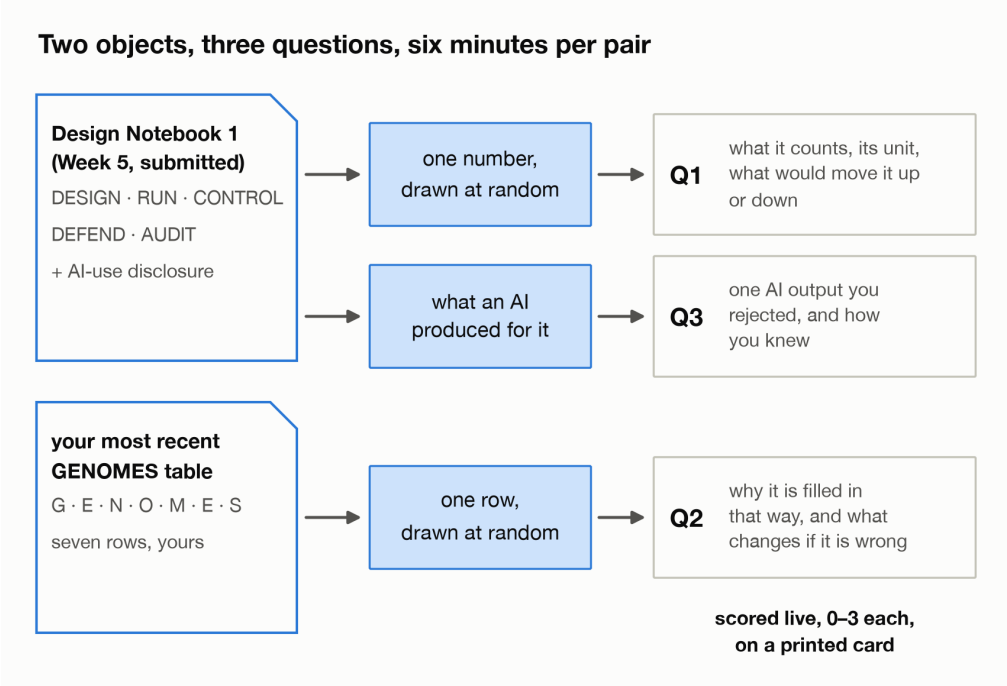
- Label transfer (etiket aktarımı) with an abstention route
  - CellTypist; popV, an ensemble that reports its uncertainty (Nat Genet 2024)
- Cell search at atlas scale: "find cells like this one" among hundreds of millions of cells
- Zero-shot cross-species transfer: carry labels from a well-studied species to a poorly annotated one
- What they share: knowledge from far more cells than any one lab will ever have
- On your sheet a learned embedding beats the baseline on one of the two metrics. You must report it.



Simulated: a label when the neighbours agree, an abstention when they do not

The micro-viva starts today

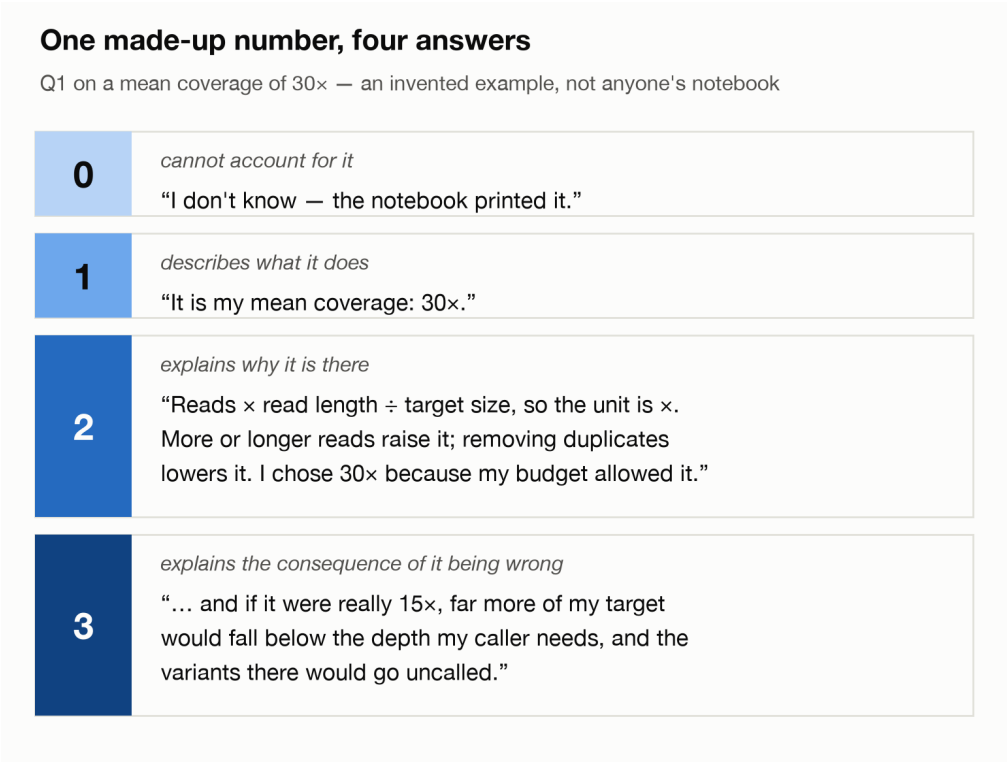
- Starts in this week's application hour, at the front desk from minute 3; runs Weeks 11–14
- Paired 6-minute slots: two students at the desk, each answering separately about their own work
- Every viva opens the same two objects: your Design Notebook 1 and your most recent GENOMES table
- The same two objects for the whole cohort — that is what makes it fair
- Your slot is on the published list on EgeDers. When your name is called, go; your worksheet waits.
- The micro-viva (mini sözlü sınav) is 18% of the grade and a hurdle (baraj) — rules two slides on



Two objects, three questions, six minutes per pair

The three questions, verbatim

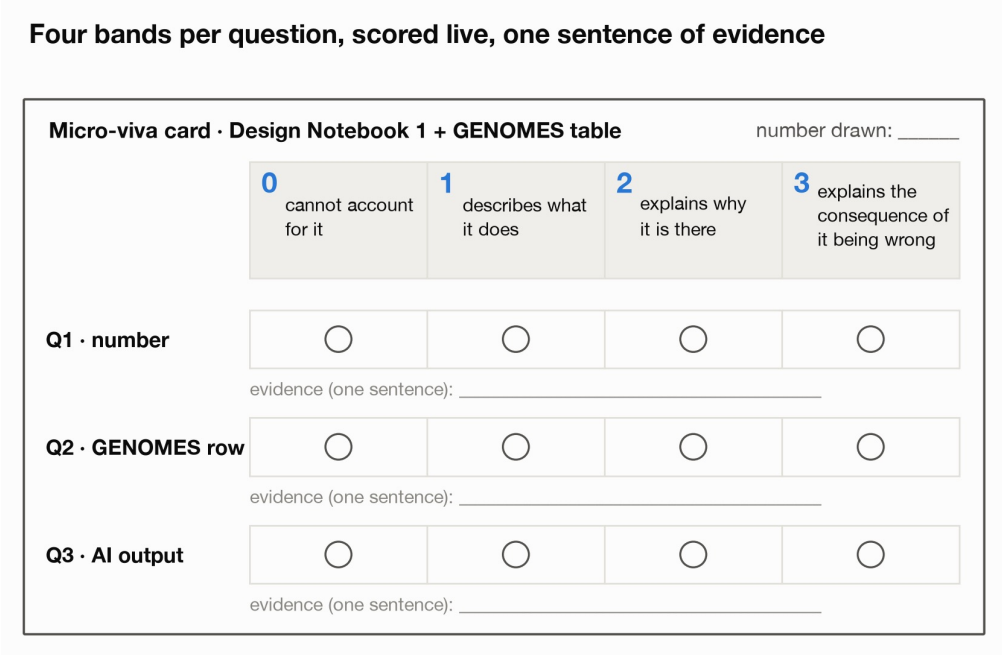
- Q1 "Here is one number your notebook recorded — tell me what it counts or measures,
  - name its unit or scale,
  - and name one thing that would have made it larger and one that would have made it smaller."
- Q2 "Read me this row of your GENOMES table: why is it filled in the way it is,
  - and what would change if it were wrong?"
- Q3 "Name one thing an AI produced for this notebook that you rejected, and how you knew."



A made-up example: the same number accounted for at each of the four bands

How it is scored: four bands, live, on a card

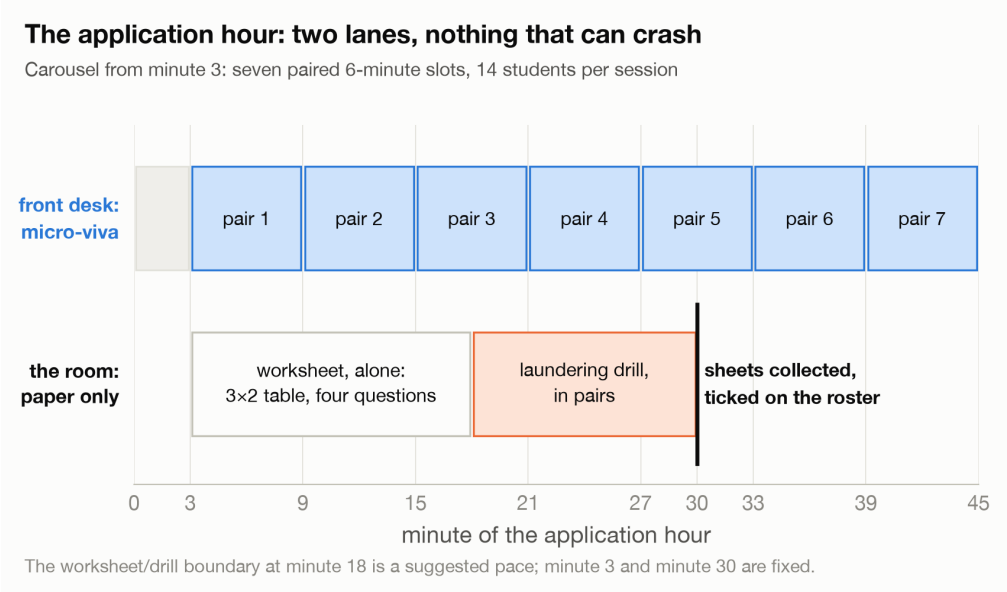
- Each question scored live, 0–3, on a printed card. Nothing is marked afterwards.
- 0 cannot account for it · 1 describes what it does · 2 explains why it is there
- 3 explains the consequence of it being wrong
- One recorded sentence of evidence per question: that is your appeal trail
- Hurdle: below 50/100 on the viva caps the course grade at the pass boundary
- Below 50: exactly one re-defence (yeniden savunma) in bütünleme week, same two objects, capped at 60/100



The scoring card: three questions, four bands, one evidence line each

The application hour: paper, pairs, and the desk

- Paper only: no laptop, no notebook, no A.finish(). One bilingual A4 worksheet each.
- One 3×2 table for the whole room: three embeddings × two metrics, run once
- Four questions, alone. Then the laundering drill, in pairs: the same table, baseline row deleted.
- Write the sentence a reader would wrongly conclude, and the one question that would have caught it
- Sheets collected at minute 30, ticked on the roster. Not scored — the sheet is your receipt.
- The carousel runs from minute 3. If your name is called, go; your sheet waits for you.



The application hour: the desk lane and the room lane